LESSON 4.1

Understand Percent

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can explain a percent as a rate per 100, model it on a decimal grid or tape diagram, and interpret percents greater than 100%.

WHY THIS MATTERS The arcade posts every result as a percent of the weekly goal. A percent compares a number to 100 — so 40% means 40 out of every 100. Some players land under the goal, and some blow past it: a score of 150% means one and a half times the whole goal. Your job is to read and model each result as a comparison to 100.

REMEMBER

Understand Percent

Understanding Percent. Formula: Percent = Parts per 100 = Ratio of a number to 100 Percent means per hundred. Any part-to-whole comparison becomes a percent by rewriting it as an equivalent ratio whose second term is 100. A decimal grid makes that visible: 100 squares, one for each percent.

1. Percent means 'per hundred' (% = /100).
2. On a 100-grid, each square represents 1% = 0.01 = 1/100.
3. Fraction to Percent: Scale fraction to a denominator of 100

Words: **Percent** ***(Porcentaje)*** · **Per hundred** ***(Por cada cien)*** · **Decimal grid** ***(Cuadrícula decimal)*** · **Part-to-whole** ***(Parte a todo)*** · **Greater than 100%** ***(Mayor que 100 %)***

1 **Circle the letter of the best answer. Show how you know.**

A recipe calls for 2 pounds of apples. How many ounces is that? (1 lb = 16 oz)

(A) 8 ounces
(B) 18 ounces
(C) 32 ounces
(D) 320 ounces

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Complete the table: 1 metre = 100 centimetres, so 7 metres = ___ centimetres.

(A) 70
(B) 107
(C) 700
(D) 7,000

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which conversion factor turns minutes into hours?

(A) 1 hour / 60 minutes
(B) 1 minute / 60 hours
(C) 60 minutes / 1 hour
(D) 60 hours / 1 minute

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A ribbon is 84 inches long. How many feet is that?

(A) 7 feet
(B) 8 feet
(C) 72 feet
(D) 1,008 feet

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Shanice's cat had two kittens. The gray kitten weighs 4 ounces. The orange kitten weighs 1.5 times as much as the gray kitten. 4 out of every 10 voters chose the tiger. What percent is that?

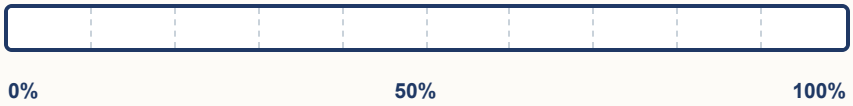
(A) 4%
(B) 10%
(C) 14%
(D) 40%

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Shanice's cat had two kittens. The gray kitten weighs 4 ounces. The orange kitten weighs 1.5 times as much as the gray kitten. On a decimal grid of 100 squares, how many squares show 23%?

(A) 2.3 squares
(B) 23 squares
(C) 77 squares
(D) 230 squares



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

EXPLAIN YOUR THINKING

A friend says "a percent can never be more than 100, because you cannot have more than all of something." How would you answer using the mascot vote and the kittens?

SENTENCE STARTER A percent compares a number to ___, so ___% just means the amount is ___ than the whole it is compared to.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each phrase under the percent it describes.

Groups: 40% 150%

_____ 40 out of 100

_____ 1.5 times the whole

_____ Less than one whole

_____ More than one whole

2 **Write the letter of the matching item on each line.**

Match each percent to its meaning.

_____ 25%

_____ 100%

_____ 150%

_____ 200%

(A) a quarter of the whole

(B) exactly one whole

(C) one and a half wholes

(D) two wholes

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Shanice's cat had two kittens. The gray kitten weighs 4 ounces. The orange kitten weighs 1.5 times as much as the gray kitten. A percent of 150% tells you the amount is…

(A) Half the whole

(B) 150 items

(C) Impossible

(D) 1.5 times the whole

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What percent is 9 out of 100?

(A) 0.9%

(B) 9%

(C) 19%

(D) 90%

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Which is exactly one whole?

(A) 1%

(B) 10%

(C) 50%

(D) 100%

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A player scored 200% of the weekly goal. That means the player scored…

(A) Half of the goal

(B) Exactly the goal

(C) Two points total

(D) Twice the goal

SHOW YOUR WORK

EXPLAIN YOUR THINKING

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